• READY FOR PRODUCTION HANDOVER | CLIENT: BALAJI AUTOMOBILES | VERIFIED SUPABASE + FCM

Project Handover & Technical Master Document

Official deployment, credentials, backend notification pipeline, and API blueprint for **Balaji Garage (Workshop Console)**, **Balaji Car (Customer App)**, and the **WhatsApp Live Shared Job Card Portal**.

🏛️ 1. System Ecosystem Overview

Balaji Garage (Console)

Flutter / Android

Workshop counter console for the owner, manager, accountant, and mechanics. Fully offline-first with immediate local persistence and automatic Supabase cloud sync.

PACKAGE NAME

```
com.balajiautomobiles.garage
```

Balaji Car (Customer)

Flutter / Android

Customer booking app allowing car owners to book service visits, track repair progress live, review estimates, inspect invoices, and find nearby workshops by GPS.

PACKAGE NAME

```
com.balajiautomobiles.customer
```

WhatsApp Live Share

React 19 / Netlify

Zero-install public web portal. A workshop sends a WhatsApp link (`/j/:token`) and the customer views their live invoice without installing an app or signing in.

BASE LINK

```
https://balajiautomobiles.in/j/
```

🔑 2. Master Credentials & Access Vault

🔒 **Confidential Master Keys:** These credentials provide administrative control over the workshop console, the Supabase PostgreSQL database, and Google Cloud services. Keep them secure.

Workshop Owner & Super Admin

Master Account

OWNER EMAIL ID

```
balajiautomobiles088@gmail.com
```

MASTER PASSWORD

```
automobiles@088*
```

PLATFORM ROLE

```
Super Admin (platform_heads) + Workshop Owner
```

WORKSHOP PHONE & UPI ID

```
9462153613 / 9269297999 | balajiauto@okicici
```

Supabase Cloud Database

PostgreSQL 15

PROJECT REFERENCE / ID

```
zlrmcrhwtfbwbxhfmygs
```

SUPABASE API BASE URL

```
https://zlrmcrhwtfbwbxhfmygs.supabase.co
```

PUBLISHABLE (ANON) KEY

```
sb_publishable_IJGQOXtE8tr7b_IgIi5cdQ_wDs6oFl7
```

DASHBOARD DIRECT LINK

```
https://supabase.com/dashboard/project/zlrmcrhwtfbwbxhfmygs
```

Google Cloud OAuth 2.0

Android & Web

WEB CLIENT ID

```
787349507364-b943fq2sf47h2eqa3chiuck7923q7gv7.apps.googleusercontent.com
```

REDIRECT URI FOR GOOGLE CLOUD CONSOLE

```
https://zlrmcrhwtfbwbxhfmygs.supabase.co/auth/v1/callback
```

KEYSTORE SHA-1 FINGERPRINT

```
83:99:64:B6:0E:36:1D:AB:E1:92:2A:F5:8D:BC:41:21:0B:20:19:41
```

Web Portals & Support

Public Links

OFFICIAL WEBSITE

```
https://balajiautomobiles.in
```

CUSTOMER APK DOWNLOAD LINK

```
https://balaji-automobiles-apps.netlify.app
```

SUPPORT WHATSAPP

```
+91 92692 97999
```

🔔 3. Backend Notification System (3-Tier Engine)

Designed to guarantee zero missed alerts across all device states. Each layer handles a distinct state where other layers cannot function.

Layer 1: In-App Banner

App Active

Driven by Supabase Realtime WebSocket channel (`garage:<id>`). Renders an instant floating banner across whatever screen is active without disrupting input.

Layer 2: Unread Feed & Bell

App Reopened

Durable records stored in `public.garage_notifications` created purely by PostgreSQL database triggers. Neither client nor owner can forge or delete them.

Layer 3: OS Tray Audio Push

App Closed

Triggered by Database Webhooks calling the `notify-garage` Edge Function to dispatch Google FCM v1 high-priority tray notifications with custom chimes.

CHANNEL ID	DESTINATION APP	AUDIO CHIME TYPE	ANDROID PRIORITY	BUSINESS EVENT TRIGGER
`garage_booking_v1`	Workshop Console	High Alert Ring	HIGH / HEADS-UP	Customer taps "Confirm Booking"
`garage_payment_v1`	Workshop Console	Payment Chime	HIGH / HEADS-UP	Counter records advance or bill settlement
`garage_reminder_v1`	Workshop Console	Standard Tone	DEFAULT	Service reminder due date reached
`garage_general_v1`	Workshop Console	Soft Chime	DEFAULT	New customer review, part inquiry
`customer_updates_v1`	Customer App	Friendly Chime	HIGH / HEADS-UP	"Vehicle Ready for Pickup", "Booking Accepted"

Edge Function Architecture: `notify-garage`

Deno TypeScript

The function runs at `https://zlrmcrhwtfbwbxhfmygs.supabase.co/functions/v1/notify-garage` and performs JWT token exchange with Google OAuth using the `FCM_SERVICE_ACCOUNT` secret.

```
// Supabase Database Webhook payload format received by notify-garage:
{
  "type": "INSERT",
  "table": "garage_notifications",
  "record": {
    "id": "notif_987654",
    "garage_id": "00000000-0000-0000-0000-000000000000",
    "kind": "booking",
    "title": "New Booking Request",
    "body": "Customer booked Periodic Service for Swift (RJ18AB1234)",
    "route": "/bookings"
  }
}
```

4. Complete API & RPC Suite

METHOD	ENDPOINT / FUNCTION	AUTHENTICATION	PARAMETERS	BUSI ACTI
POST	`/rest/v1/rpc/save_job_card`	Bearer JWT (Staff/Owner)	`{ "payload": { card, items, payments } }`	Sing trans aton job c and adva
POST	`/rest/v1/rpc/shared_card`	Public Anon Key	`{ "tok": "<192_BIT_TOKEN>" }`	Publ vehi Wha /j/: with
POST	`/rest/v1/rpc/garage_snapshot`	Bearer JWT (Staff/Owner)	`{ "gid": "<GARAGE_UUID>" }`	Fetc com filter

METHOD	ENDPOINT / FUNCTION	AUTHENTICATION	PARAMETERS	BUSI ACTI
				cach insta start
POST	/rest/v1/rpc/nearby_garages	Public Anon Key / Customer	{ "in_lat": 30.73, "in_lng": 76.77, "radius_km": 15 }	Find gara cust radiu
POST	/rest/v1/rpc/set_garage_location	Bearer JWT (Owner)	{ "gid": "...", "lat": 30.7333, "lng": 76.7794 }	Save work coor cust disc
POST	/rest/v1/rpc/read_garage_notifications	Bearer JWT (Staff/Owner)	{ "ids": ["notif_uuid_1"] }	Mark read cour hanc simu
POST	/rest/v1/rpc/punch_attendance	Bearer JWT (Staff)	{ "payload": { staff_id, lat, lng, type } }	Cloc Cloc GPS dista verif
POST	/rest/v1/rpc/issue_customer_code	Bearer JWT (Counter)	{ "cid": "<CUSTOMER_ID>" }	Gen digit code cust at cc
POST	/rest/v1/rpc/redeem_customer_code	Bearer JWT (Customer)	{ "entered": "123456" }	Cust rede to lin past histc

METHOD	ENDPOINT / FUNCTION	AUTHENTICATION	PARAMETERS	BUSI ACTI
POST	/rest/v1/rpc/register_garage_device	Bearer JWT (Staff/Owner)	{ "tok": "<FCM_TOKEN>", "plat": "android" }	Regi work hanc for F alert
POST	/rest/v1/rpc/register_customer_device	Bearer JWT (Customer)	{ "tok": "<FCM_TOKEN>", "plat": "android" }	Regi cust phor back vehi alert
POST	/functions/v1/notify-garage	Header: x-webhook-secret	Webhook payload (table, record)	Edg FCM disp trigg data INSE

Standard API Request Format (cURL Example)

PostgREST Protocol

```
curl -X POST "https://zlrmcrhwtfbwbxhfmygs.supabase.co/rest/v1/rpc/shared_card" \
  -H "apikey: sb_publishable_IJGQOXtE8tr7b_IgIi5cdQ_wDs6oFl7" \
  -H "Authorization: Bearer sb_publishable_IJGQOXtE8tr7b_IgIi5cdQ_wDs6oFl7" \
  -H "Content-Type: application/json" \
  -d '{"tok": "7a8b9c0d1e2f3a4b5c6d7e8f"}'
```

📋 5. Step-by-Step Dashboard Setup & "What to Paste Where"

1

Supabase Authentication Redirect URLs

Go to **Authentication** → **URL Configuration** → **Redirect URLs** and add all 4 items:

```
com.balajiautomobiles.garage://login-callback
com.balajiautomobiles.customer://login-callback
com.garagepro.garageproadmin://login-callback
com.garagepro.garagepro://login-callback
```

2

Supabase SQL Editor: Account & Workshop Initialization

Run this in the **SQL Editor** to ensure `balajiautomobiles088@gmail.com` has password `automobiles@088*`:

```
update auth.users
   set encrypted_password = crypt('automobiles@088*', gen_salt('bf')),
       email_confirmed_at = coalesce(email_confirmed_at, now()),
       updated_at         = now()
 where lower(email) = 'balajiautomobiles088@gmail.com';
```

3

Google Cloud Console: OAuth Web & Android Clients

In **APIs & Services** → **Credentials**:

- Add Authorised Redirect URI: `https://zlrmcrhwtfbwbxhfmygs.supabase.co/auth/v1/callback`
- Add Android Client with SHA-1: `83:99:64:B6:0E:36:1D:AB:E1:92:2A:F5:8D:BC:41:21:0B:20:19:41`

4

Supabase Edge Function: `notify-garage` Secrets

In **Edge Functions** → **Secrets**, add the following two environment variables:

SECRET NAME: WEBHOOK_SECRET

```
Your 32-character hex secret string
```

SECRET NAME: FCM_SERVICE_ACCOUNT

```
Entire contents of Firebase Service Account JSON
```

5

Supabase Database Webhooks: Booking & Update Triggers

In **Database** → **Webhooks** → **Create Hook**, add these two entries:

HOOK NAME	TARGET TABLE	EVENT	TYPE	HEADER
`notify-workshop`	`public.garage_notifications`	INSERT	Edge Function (notify-garage)	`x-webhook-secret: <WEBHOOK_SECRET>`
`notify-customer`	`public.customer_alerts`	INSERT	Edge Function (notify-garage)	`x-webhook-secret: <WEBHOOK_SECRET>`

🛡️ 6. Role-Based Access Control (RBAC)

ROLE	PLATFORM DIRECTORY	WORKSHOP SETTINGS	JOB CARDS	COUNTER SALES	PURCHASE COST & PROFIT	BILLING & LEDGER
Super Admin	✅ Full Control	✅ Global	✅ All	✅ Yes	✅ Yes	✅ Global
Owner / Admin	❌ No	✅ Full Workshop	✅ All	✅ Yes	✅ Yes	✅ Full
Manager	❌ No	❌ No	✅ All	✅ Yes	❌ Hidden	✅ Billing Only
Mechanic	❌ No	❌ No	✅ Own Cards Only	❌ No	❌ Hidden	❌ No
Counter / Store	❌ No	❌ No	✅ Create/Edit	✅ Yes	❌ Hidden	❌ No
Accountant	❌ No	❌ No	👁️ Read Only	❌ No	❌ Hidden	✅ Invoices

ROLE	PLATFORM DIRECTORY	WORKSHOP SETTINGS	JOB CARDS	COUNTER SALES	PURCHASE COST & PROFIT	BILLING & LEDGER & Reports

📑 7. GST Engine & Invoice Templates

Automated Indian GST Rules

Tax Compliance

- **Intra-State:** Splits tax 50/50 into CGST + SGST (e.g. 18% = 9% CGST + 9% SGST).
- **Inter-State:** Applies full tax slab as IGST (e.g. 18% IGST) based on customer state code.
- **Inclusive vs Exclusive Pricing:** Toggleable per job card. Inclusive calculates backwards without shifting total.
- **HSN/SAC Code Rollups:** Automatic grouping for GSTR-1 quarterly filing.
- **Auto Rounding:** Exact rupee round-off printed on a dedicated line item.

6 Built-in PDF Invoice Themes

Settings → Invoice Themes

- **Default (Original):** Standard workshop letterhead with clean table structure.
- **Default (Two Logos):** Dual branding for workshop + spare parts brand partner (Bosch, Castrol).
- **Luxury:** Dark accent typography, fine border grids, executive aesthetic.
- **Advanced GST (Tally Style):** Formal tax accountant format with complete HSN summary block.
- **Billbook:** Classic two-column counter slip layout.
- **Simple:** Minimalist single-sheet bill for routine services.

🚀 8. Release Artifacts & Client Deliverables

ARTIFACT NAME	TARGET AUDIENCE	REPOSITORY FILE LOCATION	STATUS
Balaji Admin APK v1.0.10	Workshop Owner, Staff, Counter	`release-apks/balaji-admin-1.0.10-arm64.apk`	VERIFIED PRODUCTION

ARTIFACT NAME	TARGET AUDIENCE	REPOSITORY FILE LOCATION	STATUS
Balaji Car APK v1.0.4	Customers & Vehicle Owners	`release-apks/balaji-automobiles-1.0.4-arm64.apk`	VERIFIED PRODUCTION
Admin Walkthrough Video	Workshop Staff Training	`qa-evidence/video/admin-hindi-guide.mp4`	4:35 HINGLISH NARRATION
Customer App Walkthrough	Customer Onboarding	`qa-evidence/video/user-hindi-guide.mp4`	1:30 HINGLISH NARRATION
Public Web Portal	Public & WhatsApp Link Viewers	`https://balajiautomobiles.in`	ACTIVE / READY

🛠️ 9. System Health Checks & FAQs

One-Click Verification SQL Diagnostic

Run in Supabase SQL Editor

```
-- Check 1: Super Admin & Owner Account
select u.email, u.email_confirmed_at is not null as email_confirmed, p.role
  from auth.users u
  join public.profiles p on p.id = u.id
 where lower(u.email) = 'balajiautomobiles088@gmail.com';

-- Check 2: Workshop Listing & GPS Verification
select name, phone, city, pin, lat, lng, listed, verified
  from public.garages
 where lower(email) = 'balajiautomobiles088@gmail.com';

-- Check 3: Registered Push Handset Count
select count(*) as active_workshop_handsets from public.garage_devices;
select count(*) as active_customer_phones from public.customer_devices;
```

Q: Workshop not showing up in search?

Ensure GPS coordinates are set. Open the Admin Console → **Settings** → **Workshop Details** → **Use Current Location** → **Save**. A workshop without a latitude/longitude pin is excluded from customer radius searches.

Q: Background push alert didn't ring phone?

Check Android notification permissions. On Xiaomi, Vivo, or Oppo handsets, ensure the app is granted **Autostart** and **Unrestricted Battery** permissions in Android device settings.

Balaji Automobiles Workshop Management Ecosystem — Production Release 1.0.0
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